

COMP4131: Data Modelling and Analysis

Lecture 10: Support Vector Machines and Kernel Methods

Daokun Zhang

University of Nottingham Ningbo China

daokun.zhang@nottingham.edu.cn

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FAQs for Coursework 2

- Q1: Should I compare my solutions with the state-of-the-art baselines?
- A1: Comparison with the state-of-the-art is encouraged but not compulsory.
- Q2: Should I work on the classification and regression tasks simultaneously?
- A2: Please focus on only one task. You can choose one the three tasks: classification, regression and clustering.
- Q3: Can I choose a method that has not been covered by our module?
- A3: Yes. The solution methods are not limited to what we have learned. We encourage you to explore more advanced machine learning techniques to solve your problem.

- Q4: How many solution methods should I compare?
- A4: It's expected that more than 3 methods should be investigated. Generally, the more the better, but enough words and space have to be used for describing data pre-processing details, justifying the reason for choosing the used methods, showcasing and analyzing the experimental results.
- Q5: What data scale is expected?
- A5: Generally, a dataset with more than 2000 samples is favored. If the dataset you identified contains too many samples, like in million or billion scales, you can do some down-sampling to reduce the scale.
- Q6: Can I use image or textual data?
- A6: No. Please use the tabular data.

FAQs for Coursework 2

- Q7: Can I re-use the datasets of our lab sessions?
- A7: No. Please identify some new datasets.
- Q8: Can I use the same dataset with some other student?
- A8: No. Please be different to avoid the suspicion of plagiarism.

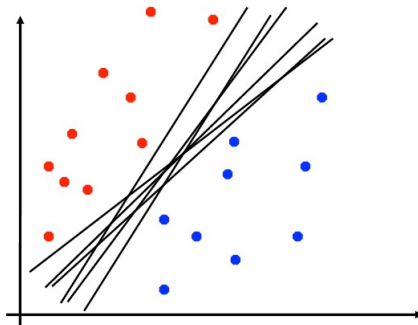
1 Support Vector Machines

2 Kernel Methods

Support Vector Machines

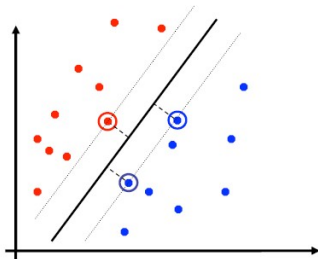
Linear Separators

- If training data is linearly separable, we can find some **linear separator** to distinguish two classes of examples.
- Which of these is **optimal**?



Support Vector Machines (SVMs)

- SVMs (Vapnik, 1990's) choose the linear separator with the largest margin.



Vladimir Vapnik

- Good according to intuition, theory, practice.
- SVMs became famous when, using images as input, it gave accuracy comparable to neural-network with hand-designed features in a handwriting recognition task.

Geometry of Linear Separators

In a high-dimensional Euclidean space, a linear separator is determined by a **hyperplane** that can be specified as the set of points given by

$$\mathbf{p} = \mathbf{a} + s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R},$$

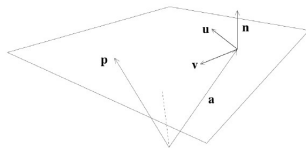
where $\mathbf{a}$ is a **vector from origin to a point in the plane**, and $\mathbf{u}$ and $\mathbf{v}$ denote **two non-parallel directions in the plane**.

Alternatively, the points can be specified as

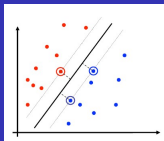
$$(\mathbf{p} - \mathbf{a}) \cdot \mathbf{n} = 0 \Leftrightarrow \mathbf{p} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n},$$

where $\mathbf{n}$ is the **norm vector**, which is also noted as $\mathbf{w}$, and the dot product $\mathbf{a} \cdot \mathbf{n}$ can be treated as a scalar to be specified, donated as the **offset** b .

So, the hyperplane can be represented as $\mathbf{w}^T \mathbf{x} + b = 0$.



Hard-SVM



Suppose we are given a set of training samples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$ that are linearly separable, the distance between the data point $\mathbf{x}_n$ with class label $y_n \in \{+1, -1\}$ and the hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ is

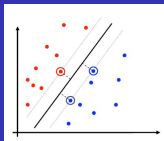
$$\frac{|\mathbf{w}^T \mathbf{x}_n + b|}{\|\mathbf{w}\|},$$

where $\|\cdot\|$ is the 2-norm of a vector.

As we are only interested in the linear separators for which all data points are correctly classified, implying that $\mathbf{w}^T \mathbf{x}_n + b \geq 0$ for $y_n = +1$ and $\mathbf{w}^T \mathbf{x}_n + b < 0$ for $y_n = -1$, i.e., $y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 0$, the distance between the data point $\mathbf{x}_n$ and the hyperplane can be rewritten as

$$\frac{y_n(\mathbf{w}^T \mathbf{x}_n + b)}{\|\mathbf{w}\|}.$$

Hard-SVM



The margin between the hyperplane and all training samples is determined by the data points closest to the hyperplane, whose value is calculated as

$$\min_n \frac{y_n(\mathbf{w}^T \mathbf{x}_n + b)}{\|\mathbf{w}\|}.$$

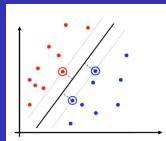
The hyperplane corresponding to the largest margin can be specified as

$$\arg \max_{\mathbf{w}, b} \left\{ \frac{1}{\|\mathbf{w}\|} \min_n [y_n (\mathbf{w}^T \mathbf{x}_n + b)] \right\},$$

where we have taken the factor $1/\|\mathbf{w}\|$ outside the optimization over n because $\mathbf{w}$ does not depend on n .

Direct solution of this optimization problem would be very complex, and so we shall convert it into an equivalent problem that is much easier to solve.

Hard-SVM



We note that if we make the rescaling $\mathbf{w} \rightarrow \kappa \mathbf{w}$ and $b \rightarrow \kappa b$, the distance between any point $\mathbf{x}_n$ to the hyperplane, given by $y_n(\mathbf{w}^T \mathbf{x}_n + b) / \|\mathbf{w}\|$, remains unchanged. We can use this freedom to set

$$y_n (\mathbf{w}^T \mathbf{x}_n + b) = 1$$

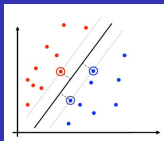
for the point that is closest to the hyperplane. In this case, all data points will satisfy the constraints

$$y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \quad n = 1, \dots, N.$$

In the case of data points for which the equality holds, the constraints are said to be **active**, whereas for the remainder they are said to be **inactive**.

By definition, there will always be **at least one active constraint**, because there will always be a closest point, and once the margin has been maximized there will be **at least two active constraints**.

Hard-SVM



The optimization problem then simply requires that we maximize $\|\mathbf{w}\|^{-1}$, which is equivalent to minimizing $\|\mathbf{w}\|^2$, and so we have to solve the optimization problem

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \quad n = 1, \dots, N.$$

The factor of $1/2$ is included for mathematical convenience. This is a quadratic programming problem in which we are trying to minimize a quadratic function subject to a set of linear inequality constraints.

The Hard-SVM formulation assumes that the training set is linearly separable, which is a rather strong assumption.

Soft-SVM can be viewed as a relaxation of the Hard-SVM rule that can be applied even if the training set is not linearly separable.

The optimization problem for Hard-SVM enforces the hard constraints $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1$ for all n .

A natural relaxation is to allow the constraint to be violated for some of the examples in the training set.

This can be modeled by introducing nonnegative slack variables, $\xi_1, \dots, \xi_N$, and replacing each constraint $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1$ by the constraint $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n$.

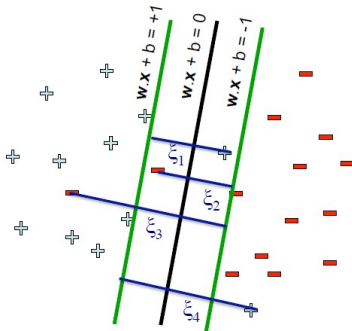
That is, ξ_n measures by how much the constraint $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1$ is being violated.

Soft-SVM

Soft-SVM jointly minimizes the norm of $\mathbf{w}$ (corresponding to [the margin](#)) and the average of ξ_n (corresponding to [the violations of the constraints](#)).

The tradeoff between the two terms is controlled by a parameter C . The Soft-SVM optimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \\ \text{s.t.} \quad & y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n \\ & \text{and } \xi_n \geq 0, \\ & n = 1, 2, \dots, N. \end{aligned}$$



Hinge Loss

For the constraint $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n$, if $y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1$ is satisfied, the minimization over ξ_n with the constraint $\xi_n \geq 0$ would make $\xi_n = 0$; otherwise, $\xi_n = 1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)$. In this sense, ξ_n can be expressed as

$$\xi_n = \max\{0, 1 - y(\mathbf{w}^T \mathbf{x}_n + b)\}.$$

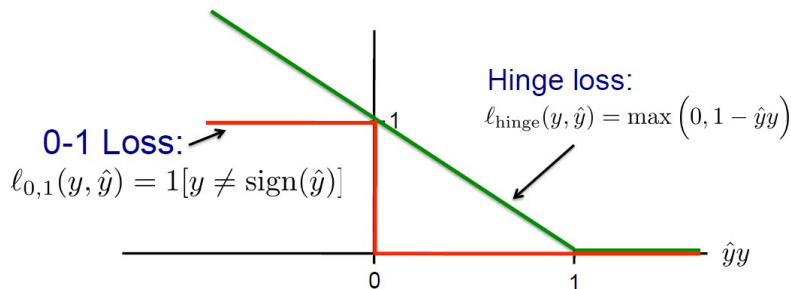
Then the Soft-SVM optimization problem can be re-formulated as

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \max\{0, 1 - y(\mathbf{w}^T \mathbf{x}_n + b)\}.$$

By denoting $\hat{y}_n = \mathbf{w}^T \mathbf{x}_n + b$, ξ_n defines the hinge loss to measure the discrepancy between the prediction $\hat{y}_n$ and the ground truth y_n :

$$\ell_{\text{hinge}}(y_n, \hat{y}_n) = \max(0, 1 - \hat{y}_n y_n).$$

Hinge Loss



Hinge loss upper bounds 0-1 loss!

It is the tightest *convex* upper bound on the 0-1 loss.

Dual Form for Hard-SVM

Given the Hard-SVM's primal optimization problem

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad s.t. \quad y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \quad n = 1, \dots, N,$$

which might be difficult to solve. We can derive its equivalent dual form, by starting with constructing a function

$$\begin{aligned} g(\mathbf{w}, b) &= \max_{\alpha \in \mathbb{R}^N: \alpha \geq 0} \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} \\ &= \begin{cases} 0 & \text{if } \forall n, y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 \\ \infty & \text{otherwise} \end{cases}. \end{aligned}$$

Dual Form for Hard-SVM

The Hard-SVM's primal optimization problem can be re-formulated as

$$\min_{\mathbf{w}, b} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + g(\mathbf{w}, b) \right\}, \text{ i.e.,}$$
$$\min_{\mathbf{w}, b} \max_{\alpha \in \mathbb{R}^N: \alpha \geq 0} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} \right\}.$$

Now suppose that we flip the order of min and max in the above equation. This can only decrease the objective value, and we have

$$\min_{\mathbf{w}, b} \max_{\alpha \in \mathbb{R}^N: \alpha \geq 0} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} \right\}$$
$$\geq \max_{\alpha \in \mathbb{R}^N: \alpha \geq 0} \min_{\mathbf{w}, b} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} \right\}.$$

Dual Form for Hard-SVM

The inequality ($\geq$) is called **weak duality**. It turns out that in our case, **strong duality** also holds; namely, the inequality holds with equality.

Therefore, the dual problem is

$$\max_{\alpha \in \mathbb{R}^N: \alpha \geq 0} \min_{\mathbf{w}, b} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} \right\}.$$

We can simplify the dual problem by noting that once α is fixed, the optimization problem with respect to $\mathbf{w}$ is unconstrained and the objective is differentiable; thus, at the optimum, the gradient equals zero:

$$\mathbf{w} - \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n = 0 \Rightarrow \mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n.$$

Dual Form for Hard-SVM

However, as $\frac{1}{2}\|\mathbf{w}\|^2 + \sum_{n=1}^N \alpha_n \{1 - y_n (\mathbf{w}^T \mathbf{x}_n + b)\}$ is a linear function of b with coefficient $-\sum_{n=1}^N \alpha_n y_n$, its minimum value would be $-\infty$ for all α except for the α satisfying

$$\sum_{n=1}^N \alpha_n y_n = 0,$$

which is a necessary condition to make sure the maximization dual problem has a valid optimal solution.

The Hard-SVM's dual optimization problem can be finally formulated as

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0. \end{aligned}$$

Dual Form for Hard-SVM

Once we find the optimal solution α^* to the dual optimization problem, how shall we convert it to the optimal solution $\mathbf{w}^*$ and b^* to the primal problem?

The optimal $\mathbf{w}^*$ is straightforward with $\mathbf{w}^* = \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n$.

The optimal b^* is a bit difficult to be derived. At the optimal solution with $\mathbf{w}^*$ and α^* having been specified, b^* should be the optimal solution to the optimization problem

$$\min_b \left\{ \frac{1}{2} \|\mathbf{w}^*\|^2 + \sum_{n=1}^N \alpha_n^* \left\{ 1 - y_n (\mathbf{w}^{*\top} \mathbf{x}_n + b) \right\} \right\}.$$

If $\alpha_n^* > 0$ for only one n , the value of $\alpha_n^* \{1 - y_n (\mathbf{w}^{*\top} \mathbf{x}_n + b)\}$ should be minimized with the optimal b^* .

As it is impossible for $1 - y_n (\mathbf{w}^{*\top} \mathbf{x}_n + b^*) < 0$ (otherwise α_n^* would be zero for the optimality), b^* should satisfy $1 - y_n (\mathbf{w}^{*\top} \mathbf{x}_n + b^*) = 0$.

Dual Form for Hard-SVM

More rigorously, it can be proved that

1. There exists at least one $\alpha_n^* > 0$, and
2. $1 - y_n (\mathbf{w}^{*T} \mathbf{x}_n + b^*) = 0$ for all $\alpha_n^* > 0$ (KKT condition).

The data point $\mathbf{x}_n$ with $\alpha_n > 0$ are termed as **support vectors**. Multiplying the left and right sides of the equations $1 - y_n (\mathbf{w}^{*T} \mathbf{x}_n + b^*) = 0$ by y_n , we can solve b^* as

$$\begin{aligned} b^* &= \frac{\sum_{n=1}^N \mathbb{I}(\alpha_n^* > 0) \cdot (y_n - \mathbf{w}^{*T} \mathbf{x}_n)}{\sum_{n=1}^N \mathbb{I}(\alpha_n^* > 0)} \\ &= \frac{\sum_{n=1}^N \mathbb{I}(\alpha_n^* > 0) \cdot \left(y_n - \sum_{m=1}^N \alpha_m^* y_m \mathbf{x}_m^T \mathbf{x}_n \right)}{\sum_{n=1}^N \mathbb{I}(\alpha_n^* > 0)}, \end{aligned}$$

where $\mathbb{I}(\alpha_n^* > 0)$ is an indicator function, whose value is 1 if $\alpha_n^* > 0$, and 0 otherwise.

Dual Form for Hard-SVM

Given the feature vector of a new test sample $\mathbf{x}$, its label $\hat{y}$ can be predicted as

$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w}^{*T} \mathbf{x} + b^* \geq 0 \\ -1 & \text{if } \mathbf{w}^{*T} \mathbf{x} + b^* < 0 \end{cases}.$$

Equivalently,

$$\hat{y} = \begin{cases} +1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n^T \mathbf{x} + b^* \geq 0 \\ -1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n^T \mathbf{x} + b^* < 0 \end{cases}.$$

Dual Form for Soft-SVM

Given the Soft-SVM's primal optimization problem

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \\ \text{s.t.} \quad & y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n \text{ and } \xi_n \geq 0, \\ & n = 1, 2, \dots, N. \end{aligned}$$

We can also derive its equivalent dual form, by starting with constructing a function

$$\begin{aligned} g(\mathbf{w}, b, \xi) &= \max_{\alpha \geq 0, \mu \geq 0} \left\{ \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\} \\ &= \begin{cases} 0 & \text{if } \forall n, y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n \text{ and } \xi_n \geq 0 \\ \infty & \text{otherwise} \end{cases}. \end{aligned}$$

Dual Form for Soft-SVM

The Soft-SVM's primal optimization problem can be re-formulated as

$$\min_{\mathbf{w}, b, \xi} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + g(\mathbf{w}, b, \xi) \right\}, \text{ i.e.,}$$

$$\min_{\mathbf{w}, b, \xi} \max_{\alpha \geq 0, \mu \geq 0} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\}.$$

Now suppose that we flip the order of min and max in the above equation. This can only decrease the objective value, and we have

$$\begin{aligned} & \min_{\mathbf{w}, b, \xi} \max_{\alpha \geq 0, \mu \geq 0} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\} \\ & \geq \\ & \max_{\alpha \geq 0, \mu \geq 0} \min_{\mathbf{w}, b, \xi} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\}. \end{aligned}$$

Dual Form for Soft-SVM

The inequality ($\geq$) is called **weak duality**. It turns out that in our case, **strong duality** also holds; namely, the inequality holds with equality. Therefore, the dual problem is

$$\max_{\alpha \geq 0, \mu \geq 0} \min_{\mathbf{w}, b, \xi} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\}.$$

We can simplify the dual problem by noting that once α and μ are fixed, the optimization problem with respect to $\mathbf{w}$ is unconstrained and the objective is differentiable; thus, at the optimum, the gradient equals zero:

$$\mathbf{w} - \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n = 0 \Rightarrow \mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n.$$

Dual Form for Soft-SVM

$\left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\}$ is a linear function of b with coefficient $-\sum_{n=1}^N \alpha_n y_n$, whose minimum value would be $-\infty$ for all α except for the α satisfying

$$\sum_{n=1}^N \alpha_n y_n = 0.$$

$\left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n + \sum_{n=1}^N \alpha_n \{1 - \xi_n - y_n (\mathbf{w}^T \mathbf{x}_n + b)\} - \sum_{n=1}^N \mu_n \xi_n \right\}$ is also a linear function of ξ_n with coefficient $C - \alpha_n - \mu_n$, whose minimum value would be $-\infty$ for all choices of α_n and μ_n except for the choice of α_n and μ_n satisfying

$$C - \alpha_n - \mu_n = 0.$$

Dual Form for Soft-SVM

The Soft-SVM's dual optimization problem can be finally formulated as

$$\begin{aligned} \max_{\alpha, \mu \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0, \\ & \mu_n \geq 0 \text{ and } C - \alpha_n - \mu_n = 0 \text{ for } n = 1, 2, \dots, N. \end{aligned}$$

Equivalently, a concise formulation is

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & C \geq \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0. \end{aligned}$$

Dual Form for Soft-SVM

Given the optimal solution α^* to the Soft-SVM's dual optimization problem, we can recover the optimal solution $\mathbf{w}^*$ and b^* to the primal problem as

$$\mathbf{w}^* = \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n,$$
$$b^* = \frac{\sum_{n=1}^N \mathbb{I}(C > \alpha_n^* > 0) \cdot \left(y_n - \sum_{m=1}^N \alpha_m^* y_m \mathbf{x}_m^T \mathbf{x}_n \right)}{\sum_{n=1}^N \mathbb{I}(C > \alpha_n^* > 0)},$$

where $\mathbb{I}(C > \alpha_n^* > 0)$ is an indicator function, whose value is 1 if $C > \alpha_n^* > 0$, and 0 otherwise.

Dual Form for Soft-SVM

Given the feature vector of a new test sample $\mathbf{x}$, its label $\hat{y}$ can be predicted as

$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w}^{*\text{T}}\mathbf{x} + b^* \geq 0 \\ -1 & \text{if } \mathbf{w}^{*\text{T}}\mathbf{x} + b^* < 0 \end{cases}.$$

Equivalently,

$$\hat{y} = \begin{cases} +1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n^{\text{T}} \mathbf{x} + b^* \geq 0 \\ -1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n \mathbf{x}_n^{\text{T}} \mathbf{x} + b^* < 0 \end{cases}.$$

Kernel Methods

Kernel SVM

Recall the Soft-SVM's dual optimization problem formulation

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & C \geq \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0, \end{aligned}$$

where the training samples' feature vectors $\mathbf{x}_n$ could be replaced by the transformed feature vectors $\phi(\mathbf{x}_n)$ through a non-linear basis function $\phi(\cdot)$:

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \phi(\mathbf{x}_n)^T \phi(\mathbf{x}_m) + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & C \geq \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0. \end{aligned}$$

Kernel Trick

The dot product $\phi(\mathbf{x}_n)^T \phi(\mathbf{x}_m)$ can be extended to any **kernel functions**.

For all $\mathbf{x}$ and $\mathbf{x}'$ in the input space $\mathcal{X}$, the function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is called a kernel function, if there exists a feature map $\varphi : \mathcal{X} \rightarrow \mathcal{V}$ that satisfies

$$k(\mathbf{x}, \mathbf{x}') = \langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle_{\mathcal{V}},$$

where $\mathcal{V}$ is a inner product space, and $\langle \cdot, \cdot \rangle_{\mathcal{V}}$ is the inner product operation defined by $\mathcal{V}$.

An alternative definition can be formulated by the **positive semidefinite (PSD)** property: For any points $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ in $\mathcal{X}$, and all choices of n real-valued coefficients $(c_1, \dots, c_N)$, the function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a kernel function, if

$$\sum_{n=1}^N \sum_{m=1}^N k(\mathbf{x}_n, \mathbf{x}_m) c_n c_m \geq 0.$$

The spanned matrix, $\mathbf{K} \in \mathbb{R}^N \times \mathbb{R}^N$ with its nm -th entry $K_{nm} = k(\mathbf{x}_n, \mathbf{x}_m)$, is called **Gram matrix**.

Many kernels can be chosen for various application scenarios

- Fisher kernel
- Polynomial kernel
- Radial basis function kernel (RBF)
- String kernels
- Graph kernels

The **Radial basis function kernel (RBF)** is also called **squared-exp** or **Gaussian** kernel, which is formulated as

$$k_{\text{SE}}(\mathbf{x}_n, \mathbf{x}_m) = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x}_n - \mathbf{x}_m\|_2^2}{2\ell^2}\right).$$

The kernel version of Soft-SVM's dual optimization problem is

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} & \left\{ -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m k(\mathbf{x}_n, \mathbf{x}_m) + \sum_{n=1}^N \alpha_n \right\} \\ \text{s.t. } & C \geq \alpha_n \geq 0 \text{ for } n = 1, 2, \dots, N \text{ and } \sum_{n=1}^N \alpha_n y_n = 0. \end{aligned}$$

With the optimal solution α^* , for a new test sample with feature vector $\mathbf{x}$, its labeled can be predicted as

$$\hat{y} = \begin{cases} +1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n k(\mathbf{x}, \mathbf{x}_n) + b^* \geq 0 \\ -1 & \text{if } \sum_{n=1}^N \alpha_n^* y_n k(\mathbf{x}, \mathbf{x}_n) + b^* < 0 \end{cases},$$

where

$$b^* = \frac{\sum_{n=1}^N \mathbb{I}(C > \alpha_n^* > 0) \cdot \left\{ y_n - \sum_{m=1}^N \alpha_m^* y_m k(\mathbf{x}_m, \mathbf{x}_n) \right\}}{\sum_{n=1}^N \mathbb{I}(C > \alpha_n^* > 0)}.$$

- Christopher M. Bishop. **Pattern Recognition and Machine Learning**. New York: Springer; 2006 Aug 17. ([Chapter 7.1 Maximum Margin Classifiers](#))
- Shai Shalev-Shwartz, and Shai Ben-David. **Understanding Machine Learning: From Theory to Algorithms**. Cambridge university press, 2014. ([Chapter 15 Support Vector Machines](#))
- https://en.wikipedia.org/wiki/Support_vector_machine

The End